Medium voltage

33 kV single core cable XLPE-AL — Halogen free



Application

Application	33 kV A.C. Electricity distribution cable.
Conductor temperature	Max. 90°C
Short circiut temperature	Max. 250°C
Installation temperature	Min. -15°C
Min. radius of bend	15 x D
Max. pullings tension	Directly on conductor 30 N/mm² of total conductor cross section.

Construction

Conductor	Circular solid aluminium conductor
Conductor screen	Extruded semi-conducting compound
Insulation	XLPE
Insulation screen	Extruded semi-conducting compound
Screen	Concentric layer of copper wires with a copper tape applied in the opposite direction.
Outer sheath	PE
Colour of sheath	Black

Technical

Standard	BS 7870
Marking of sheath	ELECTRIC CABLE 33000 V nkt cables ‘Dimension’ ‘Year’ ‘Metres’ (additional text on request)
Rated voltage	33 kV AC between phases
Standard length	Available on request
Screen sizes	25 mm², 35 mm², 50 mm² on request

Nominel area of conductor and screen	Diameter of conductor nominel	Radial thickness of insulation	Standard lenght	Outer dimension max. (D)	Approx weight	EAN-nr
mm²	mm	mm	m	mm	kg pr. km	
1X50+16	7.5	8.0	On request	38.0	1250	
1X95+25	10.7	8.0	-	43.0	1450	
1X150+25	13.3	8.0	-	46.0	1700	
1X185+50	14.8	8.0	-	47.0	2100	
1X240+50	17.0	8.0	-	49.5	2300	
1X300+50	18.9	8.0	-	52.0	2600	
1X400+50	21.4	8.0	-	54.0	3100	
1X500+50	24.2	8.0	-	55.0	3500	

Mechanical properties

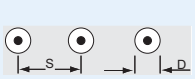
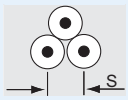
Area of conductor	mm²	50	95	150	185	240	300	400	500
Diameter of conductor, nom.	mm	7.5	10.7	13.3	14.8	17.0	18.9	21.4	24.2
Insulation thickness, nom	mm	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Diameter over isulation, nom.	mm	25.1	29.0	32.4	32.4	34.6	36.5	39.0	41.6
Area of copper screen, nom.	mm^2	16	25	25	50	50	50	50	50
Diameter over screen, nom.	mm	30.3	34.0	37.0	38.4	40.3	42.5	45.0	46.7
Thickness of sheath, nom.	mm	1.8	2.1	2.2	2.2	2.3	2.4	2.5	2.6
Diameter over sheath, nom.	mm	34.1	39.0	42.0	43.0	45.5	47.6	50.3	52.4
Diameter over sheath, max.	mm	38	43.0	46.0	47.0	49.5	52.0	54.0	55.0
weight of cable. appr.	kg/m	1.25	1.45	1.70	2.10	2.30	2.60	3.10	3.50
Radius of bend, min.	mm	540	585	630	650	680	720	750	780
Pulling tension on the cable.	kN	2.5	4.3	5.7	6.6	8.0	9.6	12.1	14.6

Electrical properties

Area of conductor	mm²	50	95	150	185	240	300	400	500
Resistance, DC, at 20°C, max.	Ω/km	0.641	0.320	0.206	0.164	0.125	0.100	0.078	0.0605
Capacitance, max.	µF/km	0.14	0.18	0.20	0.22	0.24	0.26	0.29	0.32
Short circuit rating for 1 sec. Conductor with initial temperature 90°C and final temperature 250°C	kA	4.8	9.0	14.2	17.5	22.6	28.2	37.6	47.0
Short curcuit rating for 1 sec. Screen with final screen temperature 300°C	kA	3.2	5.0	5.0	10.0	10.0	10.0	10.0	10.0

Current carrying capacity

G)Continuous current carrying capacity for maximum conductor temperature 90°C: Direct in ground at 15°C, dept 1,0 m, thermal resistivity 1,2° Km/W.
A)Continuous current carrying capacity for maximum conductor temperature 90°C: In free air at. 25°C.

Compacted conductor area	mm²	50	95	150	185	240	300	400	500
Cable formation									
 									
G)									
Screens bonded at both ends									
trefoil formation, close together S = D	A	180	250	315	350	415	465	530	600
flat formation, close together S = D	A	180	250	315	350	410	460	520	580
flat formaton, free distance between cable S = D + 70 mm	A	200	260	330	365	420	470	530	630
Screens bonded at a single point									
trefoil formation, close together S = D	A	180	250	320	350	415	470	535	610
flat formation, close together S = D	A	180	250	320	350	415	470	535	610
flat formaton, free distance between cable S = D + 70 mm	A	200	265	340	380	445	505	575	655
A)									
Screens bonded at both ends									
trefoil formation, close together S = D	A	205	300	390	445	525	600	700	805
flat formation, close together S = D	A	210	310	405	460	540	615	715	810
flat formaton, free distance between cable S = D + 70 mm	A	215	330	425	480	560	630	730	800
Screens bonded at a single point									
trefoil formation, close together S = D	A	205	300	395	450	530	610	710	825
flat formation, close together S = D	A	210	315	410	470	555	635	745	860
flat formaton, free distance between cable S = D + 70 mm	A	220	340	440	505	595	685	800	930
Reactance at 50 Hz									
trefoil formation, close together S = D	Ω/km	0.15	0.13	0.13	0.12	0.12	0.11	0.11	0.11
flat formation, close together S = D	Ω/km	0.16	0.15	0.14	0.14	0.13	0.13	0.12	0.12
flat formaton, free distance between cable S = D + 70 mm	Ω/km	0.23	0.21	0.20	0.19	0.18	0.18	0.17	0.16